

INDIA DATA RESEARCH ACADEMY

DATA SCIENCE & AI TRAINING PROGRAM

CAPSTONE PROJECT

RESEARCH-ARTICLE STYLE DATA SCIENCE PROJECT

Complete Guide to Planning, Analysing, Modelling, Writing and Submitting Your Project

SUBMISSION DEADLINE: 20 September 2026. You have a 20-day project window to complete and submit your capstone.

Project Report Submitted in fulfillment of the requirements for the

Data Science & AI Training Program by IDRA

Department of Data Science and AI

This guide is designed to help you produce a professional, research-style capstone report that combines the data analysis, statistics, visualization and machine learning skills covered in the training program.

1. What This Capstone Is

The capstone is the final integrated project of the training phase. You will take a real-world dataset, understand its structure and business or research context, clean and preprocess it, conduct exploratory data analysis, apply statistical reasoning, build a suitable machine learning model to predict an outcome, evaluate the model, and communicate your findings in a professional research-article style report. The official coursework places the capstone after Python, NumPy/Pandas, data cleaning, EDA, visualization, statistics and the introductory machine learning workflow.

The goal is not to use the maximum number of techniques. The goal is to make sensible analytical decisions, explain why you made them, support your conclusions with evidence, and present the work professionally.

What you are expected to produce

- A research-style project report in PDF format.
- A complete Jupyter/Google Colab notebook containing your code and reproducible analysis.
- A cleaned/preprocessed dataset or other supporting files when appropriate.
- A professional Google Drive folder containing the required files, from which you will submit one public shareable link on the LMS.

Project timeline

Item	Requirement
Project duration	20 days
Final submission date	20 September 2026
Recommended work style	Work continuously in the same notebook and report. Do not wait until the final week to begin writing.
Final format	PDF report + supporting notebook/files in Google Drive

How your capstone should read

Think of your report as a short research article written for a technically literate reader who does not know your dataset. A reader should be able to understand the problem, dataset, methodology, evidence, model performance, limitations and final conclusions without having to run your code. Your notebook contains the detailed computational record; the PDF report communicates the important story.

2. Choose Your Project

Choose ONE project title from the list below. A corresponding dataset will be provided for each project in the LMS. You are expected to use the dataset assigned to your selected project unless you choose the independent-project option described below.

# / PROJECT TITLE	PRIMARY FOCUS	MODEL TYPE / TARGET
1. Predictive Analytics for Healthcare: Forecasting Patient Length of Stay in Urban Hospitals	Patient characteristics, admission information, hospital operations and stay duration	Regression Target: length of stay
2. The Impact of Socio-Economic Factors on Academic Performance: A Comprehensive Analysis of Student Outcomes	Student background, learning environment, socio-economic factors and academic performance	Regression and/or classification Target: score/outcome
3. Optimizing Supply Chain Logistics: A Data-Driven Approach to Inventory Management and Demand Forecasting	Inventory levels, sales, product movement, suppliers and demand patterns	Regression/forecasting Target: demand or sales
4. Predicting Customer Churn: A Data-Driven Analysis of Subscription Customer Behaviour	Customer usage, plans, tenure, support interactions and churn behaviour	Classification Target: churn
5. Credit Risk Assessment: Predicting Loan Default Using Customer and Loan Characteristics	Borrower profiles, loan characteristics, financial indicators and repayment outcomes	Classification Target: default
6. Retail Intelligence: Analysing Sales Drivers and Predicting Product Demand	Products, customers, sales, discounts, stores and time-based demand patterns	Regression Target: sales/demand
7. Employee Attrition and Workforce Analytics: Identifying Drivers of Employee Turnover	Employee demographics, role, compensation, satisfaction and employment outcomes	Classification Target: attrition
8. Urban Air Quality Analysis: Understanding Environmental Drivers and Predicting Air Quality	Pollutants, weather conditions, locations and air-quality measurements	Regression Target: AQI or pollutant level
9. House Price Prediction: A Data-Driven Study of the Factors Influencing Residential Property Values	Property characteristics, location, size, amenities and sale prices	Regression Target: house price

# / PROJECT TITLE	PRIMARY FOCUS	MODEL TYPE / TARGET
10. Online Shopping Behaviour: Analysing Customer Activity and Predicting Purchase Intent	Sessions, traffic sources, browsing behaviour, customer attributes and purchase outcomes	Classification Target: purchase intent

Dataset access: The dataset corresponding to each listed project will be available in the LMS. Check the project instructions for the exact file assigned to your chosen title.

Independent project option

You may propose a project of your own choice instead of selecting one of the ten titles above. Your topic must be relevant to data science, must contain enough data for meaningful analysis, and must support a clear prediction or analytical objective. You may use your own dataset or a suitable public dataset. Your project must still follow the same report structure and must use only techniques that are appropriate to the problem and aligned with the skills covered in the program.

Important: Do not choose a topic simply because the dataset looks interesting. Choose a problem that has a clear question, useful variables, a meaningful target for prediction where appropriate, and enough observations to support analysis.

3. Required Report Structure

Use the following order. You may adjust the wording of section headings slightly to match your project, but do not remove the core analytical sections.

Section	What belongs here
1. Title Page	Project title, your name, institute, roll/enrollment details, programme and supervisor/institution details if required. Use a clean academic layout.
2. Abstract	A concise summary of the problem, dataset, methods, major findings, model and most important conclusion. Write this after completing the project.
3. Keywords	Add 4-6 keywords such as EDA, data preprocessing, regression, classification, predictive analytics.
4. Introduction	Explain the real-world problem, why it matters, the context of the dataset, and what the project aims to discover or predict.
5. Objectives / Research Questions	State clear objectives. Convert broad goals into specific analytical questions. Include a primary prediction question.
6. Dataset Description	Explain the source, number of rows and columns, variable types, important fields, target variable, time period or population, and any important limitations of the dataset.
7. Data Cleaning	Document missing values, duplicates, inconsistent values, incorrect data types, invalid values and how each issue was handled.
8. Data Preprocessing	Explain feature selection, encoding, scaling, outlier treatment, train-test preparation and any transformations used.
9. Exploratory Data Analysis	Use summary statistics, distributions, group comparisons, relationships and correlations to understand the data and guide the modelling decisions.
10. Data Visualisation	Present only the most useful charts. Each chart should

Section	What belongs here
	answer a question or support a finding.
11. Statistical Analysis	Use relevant statistical measures or concepts learned in the programme to support the analysis. Do not add statistical tests that you were not taught unless you clearly understand and can explain them.
12. Machine Learning Methodology	Explain whether the task is regression or classification, how features/target were selected, train-test split, preprocessing, model choice and training process.
13. Model Results & Evaluation	Present predictions and appropriate evaluation metrics. For classification, include a confusion matrix and accuracy, precision, recall and F1-score. For regression, report the regression metrics covered in your sessions and explain what they mean.
14. Discussion	Interpret the findings. Explain what the EDA and model results mean in the context of the original problem. Avoid simply repeating numbers.
15. Limitations	Identify weaknesses such as missing variables, data quality issues, sample limitations, possible bias, model limitations or lack of causal evidence.
16. Conclusion & Recommendations	Summarize the answer to the project question and, where appropriate, provide practical recommendations based on the evidence.
17. References	List datasets, websites, papers, documentation and other sources used. Use one consistent citation style.
18. Appendix	Optional supporting material such as selected code excerpts, additional tables, supplementary charts, feature dictionaries or links to the notebook.

4. How to Write Each Section

4.1 Title Page

Follow the style of the sample project covers provided to you: a strong research-style title, a statement that it is a project report submitted as part of the IDRA programme, your name and institute, relevant registration details, supervisor/institution details if required, and the India Data Research Academy identification. Keep the title specific and informative. Avoid titles such as “My Data Science Project”.

Good title pattern: Topic/context + analytical purpose + method or target. Example: “Predictive Analytics for Healthcare: Utilizing Regression Models to Forecast Patient Length of Stay in Urban Hospitals.”

4.2 Abstract

Write approximately 150-250 words. State the problem, dataset, key preprocessing, analytical approach, machine learning task, headline model result, and the most important conclusion. Do not place citations, code, or detailed tables in the abstract.

4.3 Introduction

- Start with the real-world problem, not with Python code.
- Explain why the problem matters and who could benefit from understanding it.
- Briefly describe what your project will investigate and predict.
- End the introduction with the scope of the project and the key objectives.

4.4 Objectives and Research Questions

Write 3-6 objectives. Make each one measurable. For example: “Identify the variables most strongly associated with customer churn”; “Build a classification model to predict churn”; “Evaluate the model using accuracy, precision, recall and F1-score.”

5. Dataset Description, Cleaning and Preprocessing

5.1 Describe the dataset

- State the source or origin of the dataset.
- Report the number of observations and variables.
- Create a concise data dictionary for important columns: name, meaning, type and role.
- Identify the target variable you intend to predict.
- Separate predictor/features from the target conceptually before modelling.

5.2 Document data quality issues

Issue	What to report
Missing values	Show which columns contain missing values, how many, and how you handled them.
Duplicate records	Show how duplicates were detected and whether they were removed.
Data types	Explain conversions such as string to numeric or string to datetime.
Inconsistent values	Document issues such as inconsistent spelling/capitalisation or category labels.
Outliers	Identify important numerical outliers and explain whether you removed, capped or retained them.
Validation	After cleaning, demonstrate that the problems were actually resolved.

5.3 Preprocessing

Use preprocessing only where it is needed by your model. Explain categorical encoding, numerical scaling, feature creation and any other transformation. When you scale or otherwise learn transformation parameters, use the training data to fit those transformations and then apply them to the test data. This avoids data leakage.

Do not clean blindly. Every major cleaning decision should have a reason. For example, dropping a large number of rows simply because they contain a missing value may introduce bias or unnecessarily reduce your sample.

6. Exploratory Data Analysis and Statistical Analysis

6.1 EDA sequence

1. Start with a dataset overview: shape, columns, data types and basic descriptive statistics.
2. Study the distributions of important numerical variables using appropriate plots and statistics.
3. Study important categorical variables using counts, proportions or comparison charts.
4. Compare groups that matter to your problem, such as regions, customer types, departments or product categories.
5. Investigate relationships among important numerical variables and examine correlation where appropriate.
6. Use what you learned from EDA to justify your feature selection and modelling decisions.

6.2 Do not turn EDA into a chart dump

A professional report does not contain twenty charts simply because twenty charts can be generated. Select charts that help answer your research questions. When a chart does not support a conclusion, move it to the appendix or leave it out.

For every important result, ask: What do I observe? Why might it matter? Does it answer one of my questions? What decision or modelling choice does it support?

6.3 Statistical analysis

Use appropriate statistics such as mean, median, mode, standard deviation, variance, IQR, group summaries and correlation. Be careful with interpretation. A strong correlation does not by itself prove causation. Where distributions are skewed or affected by outliers, explain whether the median provides a more representative centre than the mean.

7. How to Add Graphs, Tables and Results

7.1 Figures / graphs

Export or copy high-quality charts from your notebook into the report. Do not paste a screenshot of your entire notebook screen. Crop figures so that the graph, labels and legend are readable.

Element	Requirement
Figure number	Use sequential numbering: Figure 1, Figure 2, Figure 3, etc.
Figure title/caption	Write a short descriptive caption below the figure.
Chart content	Keep labels readable. Include units where relevant.
Interpretation	Immediately explain what the figure shows and the important insight it supports.
Source note	If a visual uses an external source rather than your own analysis, cite it.

Example figure explanation

Figure 4. Monthly Customer Churn Rate The figure shows that churn is consistently higher in the first three months of the customer lifecycle. This suggests that early-stage engagement may be an important area for retention efforts. The model therefore includes tenure-related variables as potential predictors.

7.2 Tables

Use tables for exact values that are difficult to read from a chart, such as model metrics, feature summaries or data-quality results. Give every important table a number and title. Do not paste raw DataFrame dumps with dozens of columns into the main report.

7.3 Model result presentation

For regression, report the evaluation metric values you learned and explain whether the error is acceptable for your problem. For classification, include the confusion matrix and a compact table containing accuracy, precision, recall and F1-score. Then explain the meaning of false positives and false negatives in the context of your project.

8. How to Add Code to the Report

Your complete code should remain in the submitted Jupyter/Google Colab notebook. The PDF report should contain only the code that is useful for understanding important analytical or modelling decisions.

What to include in the PDF

- Short, relevant code snippets for important cleaning steps, feature engineering, model training or evaluation.
- The output or result produced by that code when the result is important to the argument.
- A plain-language explanation of what the code does and why it was needed.

What not to do

- Do not paste the entire notebook into the PDF.
- Do not include repetitive code such as ten nearly identical plotting commands unless it adds value.
- Do not paste unreadable screenshots of code.
- Do not show code without explaining the analytical purpose behind it.

Code formatting tip: Use a monospace font, a small readable size, and a light background. Keep long code lines wrapped. Number or label especially important snippets if they are referenced in the text.

The notebook is your reproducibility record

The notebook should run from beginning to end in the correct order. Start with imports and data loading, then cleaning, preprocessing, analysis, modelling, prediction and evaluation. Keep the notebook organised with Markdown headings and short explanations between code blocks.

9. Machine Learning Section

9.1 Define the prediction problem

Clearly state whether your task is regression or classification. For regression, your target should be continuous, such as price, revenue, length of stay or demand. For classification, your target should represent a class such as Yes/No, Pass/Fail, Churn/Stay or Default/No Default.

9.2 Feature and target preparation

- Identify X, the features used for prediction, and y, the target.
- Remove or exclude variables that directly reveal the target or contain information that would only be known after the outcome occurs.
- Encode categorical features when required.
- Scale numerical features when required by the selected model or workflow.
- Keep the train-test split separate from any information used to fit preprocessing transformations.

9.3 Train the model

Use the Scikit-learn workflow taught in class. Record the model selected, important parameters, feature set and split used. Your report should make it possible for another student to understand how you trained the model without seeing every line of code.

9.4 Evaluate the model

Task	What to present
Regression	Report the regression metrics covered in class. Explain what lower/higher values mean and what the result implies about prediction error.
Classification	Report accuracy, precision, recall and F1-score, plus a confusion matrix. Explain which error type matters most for the specific application.
Overfitting/underfitting	Compare training and testing performance. Discuss whether the gap suggests overfitting, underfitting or reasonable generalisation.

Do not claim that a model is “good” only because the accuracy is high. Discuss the data balance, metric selection and the cost of different errors. A model should be judged in the context of the problem.

10. Discussion, Limitations and Recommendations

10.1 Discussion

This section connects everything. Explain how the EDA findings relate to the model, whether your original expectations were supported, which variables appeared important, and where the model succeeded or struggled. Discuss results rather than repeating the Results section word-for-word.

10.2 Limitations

- Dataset size and representativeness
- Missing or imperfect variables
- Potential data collection bias
- Outliers or unusual cases
- Limited model choice or limited training time
- Correlation versus causation
- Potential generalisation problems when applying the model to new data

10.3 Recommendations

Provide 3-5 recommendations that are genuinely supported by your findings. Recommendations should be actionable and connected to the original problem. Avoid generic statements such as “the company should use data analytics more.”

Recommendation formula: Finding -> evidence -> implication -> recommended action. Example: “Customers with short tenure show the highest churn rate; therefore, an early-stage retention programme should prioritise the first three months.”

11. References, Appendix and Academic Presentation

11.1 References

Cite the dataset source and any external material you used. Keep one consistent citation style throughout the report. Do not include a reference list containing sources that were never used.

11.2 Appendix

Use the appendix for supporting material that is useful but too detailed for the main story. Examples include full data dictionaries, additional charts, supplementary result tables, selected code snippets and extended model output.

11.3 Recommended report formatting

Element	Recommended standard
Paper	A4 or equivalent standard report size.
Main font	A clear professional font such as Aptos, Calibri, Arial or Times New Roman.
Body size	Approximately 10.5-12 pt.
Headings	Consistent numbered hierarchy.
Margins	Approximately 2-2.5 cm.
Figures	Centered, readable and numbered.
Tables	Numbered, titled and referenced in the text.
File naming	Surname_Name_Capstone_2026.pdf and Surname_Name_Capstone_Notebook.ipynb

Professional standard: Read the PDF as if you were sending it to a company manager or research supervisor. Remove spelling errors, broken tables, cut-off plots, empty pages, raw code dumps and unexplained outputs before submission.

12. Recommended 20-Day Work Plan

DAYS	STAGE	WHAT TO COMPLETE
Days 1-2	Choose topic and dataset	Understand the project question, inspect the dataset, identify possible target variables and define objectives.
Days 3-4	Dataset understanding	Explore dimensions, columns, types, missingness, duplicates and initial data quality.
Days 5-6	Cleaning	Handle missing values, duplicates, inconsistent values and data types. Record every important decision.
Days 7-8	Preprocessing	Handle outliers where justified, encode categorical data, scale/transform features where required and prepare modelling variables.
Days 9-11	EDA	Perform descriptive analysis, distributions, group comparisons, relationships and correlation analysis.
Days 12-13	Visualisation	Create and refine the most meaningful charts. Write the interpretation of each key figure.
Days 14-16	Machine learning	Build the regression or classification model, generate predictions and evaluate performance.
Day 17	Model interpretation	Investigate errors, training vs testing performance and overfitting/underfitting.
Days 18-19	Report writing	Finish the report, figures, tables, discussion, limitations, recommendations and references.
Day 20	Final QA and submission	Run the notebook, proofread the PDF, check all files and links, upload to Google Drive, test access and submit the LMS link.

Weekly-style checkpoints for yourself

- By Day 6, you should know exactly what your target variable is and have a cleaned dataset.
- By Day 11, you should have meaningful EDA results and a clear story emerging from the data.
- By Day 16, you should have a trained and evaluated model.
- By Day 19, the report should be complete. Day 20 should be quality control, not the first writing day.

13. Final Submission: Google Drive + LMS

Your submission to the LMS will be a public, viewable Google Drive link. The link should lead to a folder containing your final PDF report and all supporting files required to verify your work.

Step 1: Prepare your final files

- Final project report in PDF format.
- Final Jupyter/Google Colab notebook (.ipynb).
- Cleaned/preprocessed dataset or supporting data files, if applicable.
- Any additional files that are necessary to reproduce or understand your project.

Step 2: Create a Google Drive folder

1. Create one folder and name it clearly, for example: “IDRA Capstone - YourName - ProjectTitle”.
2. Upload the final PDF, notebook and required supporting files into this folder.
3. Remove draft versions so that the folder clearly contains the final submission.

Step 3: Set the sharing permission

1. Right-click the Google Drive folder and choose Share.
2. Under General access, select Anyone with the link.
3. Set the permission to Viewer unless your instructor specifically requests another permission level.
4. Copy the folder link.

Test your link before submitting. Open the link in an incognito/private browser window or another account. If the folder cannot be opened without your login, the submission is not ready.

Step 4: Submit on the LMS

1. Open the IDRA LMS.
2. Go to Assignments.
3. Open the Capstone Project assignment.
4. Paste the public Google Drive folder link into the submission field.
5. Submit/Save the assignment and confirm that the submission has been recorded.

14. Final Submission Checklist

Before you submit, check every item below. A technically correct analysis can still lose quality marks if the presentation, reproducibility or submission process is incomplete.

- ☐ **01** Project title is specific and matches the selected topic/dataset.
- ☐ **02** Abstract summarises the problem, approach, major findings and conclusion.
- ☐ **03** Dataset source, dimensions and important variables are documented.
- ☐ **04** Missing values, duplicates, data types, inconsistencies and relevant outliers were checked and addressed.
- ☐ **05** Cleaning and preprocessing decisions are explained, not just performed silently.
- ☐ **06** EDA includes meaningful statistics and comparisons.
- ☐ **07** Visualisations are readable, numbered and interpreted.
- ☐ **08** The prediction target is clearly defined.
- ☐ **09** Features and target are separated correctly and no obvious target leakage is present.
- ☐ **10** Train-test split is used appropriately.
- ☐ **11** Model choice is justified.
- ☐ **12** Regression or classification evaluation is complete and correctly interpreted.
- ☐ **13** Overfitting/underfitting is discussed using training and testing performance.
- ☐ **14** Key findings are evidence-based.
- ☐ **15** Limitations are acknowledged.
- ☐ **16** Recommendations are specific and connected to findings.
- ☐ **17** References are included and consistent.
- ☐ **18** Notebook runs in order and contains useful Markdown explanations.
- ☐ **19** PDF has been proofread and contains no broken tables, clipped charts or unreadable code.
- ☐ **20** Google Drive folder contains only the final/supporting files.
- ☐ **21** Google Drive link works for an external viewer.
- ☐ **22** LMS submission contains the correct public link.

Academic integrity and responsible use of tools

Your report and analysis must represent your own work. You may use documentation, tutorials and responsible assistance for learning or debugging, but you must understand every part of the submitted notebook and be able to explain your results. Do not fabricate results, invent data quality findings, or copy a project report from another person. Any external dataset, image, figure or source must be appropriately acknowledged.

Final principle: A strong capstone is a coherent story: a real problem, reliable data, justified preparation, evidence-based analysis, a properly evaluated model, honest limitations and clear conclusions.